Bluetooth RC Car Using Arduino UNO R4

Project Title & Mission

Goal

Design and implement a wireless remote-controlled vehicle.

Core Platform

Arduino UNO R4 WiFi.

Primary Features

Bluetooth Low Energy (BLE) control, mobile app integration, and dual-motor propulsion.

Project Overview

Wireless Control:

Leverages BLE for low-latency communication with mobile devices.

Movement:

Driven by two DC motors via an L298N dual H-bridge motor driver.

Features:

- Adjustable motor speed.
- Active safety using ultrasonic sensors.
- Visual feedback via an integrated LED matrix.

Hardware Components

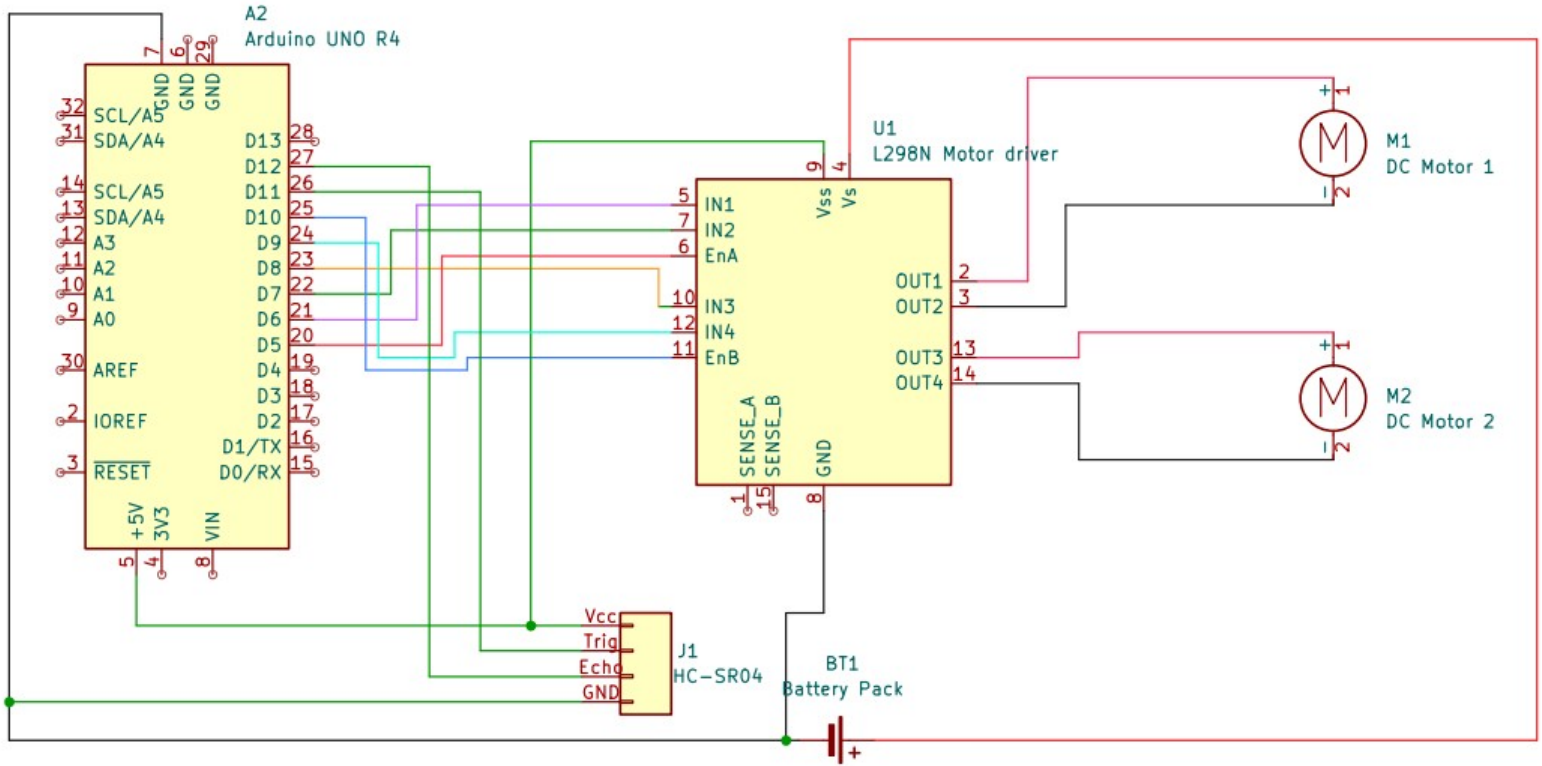
Microcontroller: Arduino UNO R4 WiFi, featuring an ESP32-S3-Mini-1 co-processor for connectivity.

Motor Driver: L298N Dual H-Bridge for handling high-current DC motors.

Sensing: HC-SR04 Ultrasonic Sensor for obstacle detection.

Power: 6 AA battery pack.

Schematic


A2
Arduino UNO R4
SCL/A5
SDA/A4
SCL/A5
SDA/A4
A3
A2
A1
A0
AREF
IOREF
RESET
+5V
3V3
VIN
GND
GND
GND
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1/TX
D0/RX
U1
L298N Motor driver
IN1
IN2
EnA
IN3
IN4
EnB
Vss
Vs
OUT1
OUT2
OUT3
OUT4
SENSE_A
SENSE_B
GND
M1
DC Motor 1
M2
DC Motor 2
Vcc
Trig
Echo
GND
J1
HC-SR04
BT1
Battery Pack

Software Architecture

Environment

Developed in C++ using the Arduino IDE.

Modular Design

Software is split into independent files for improved readability and extensibility.

Key Modules

- **Main.ino**: Orchestrates the system loop.
- **motor_control**: Logic for motor movement.
- **ble_control**: Handles Bluetooth communication.
- **sensor_control**: Manages ultrasonic data

Motor Control & Logic

Abstraction

The system uses high-level "direction" functions rather than direct pin manipulation.

Speed Control

Managed via analogWrite on ENA and ENB pins

Motion truth table

Direction	IN1	IN2	IN3	IN4	Logic
Forward	HIGH	LOW	HIGH	LOW	Both motors spin forward.
Backward	LOW	HIGH	LOW	HIGH	Polarity is reversed for both motors.
Left Turn	HIGH	LOW	LOW	HIGH	Right motor forward, Left motor backward.
Right Turn	LOW	HIGH	HIGH	LOW	Left motor forward, Right motor backward.
Stop	LOW	LOW	LOW	LOW	All coils de-energized.

BLE Communication

Architecture

Arduino acts as the Peripheral (Server), while the phone acts as the Central (Client).

Service & Characteristic

- **Service**: "CarControlService".
- **Characteristic**: "CommandCharacteristic" receives single-character commands (e.g., 'F', 'B', 'S').
- **Library**: Uses the official ArduinoBLE library.

Mobile App Interface

Application

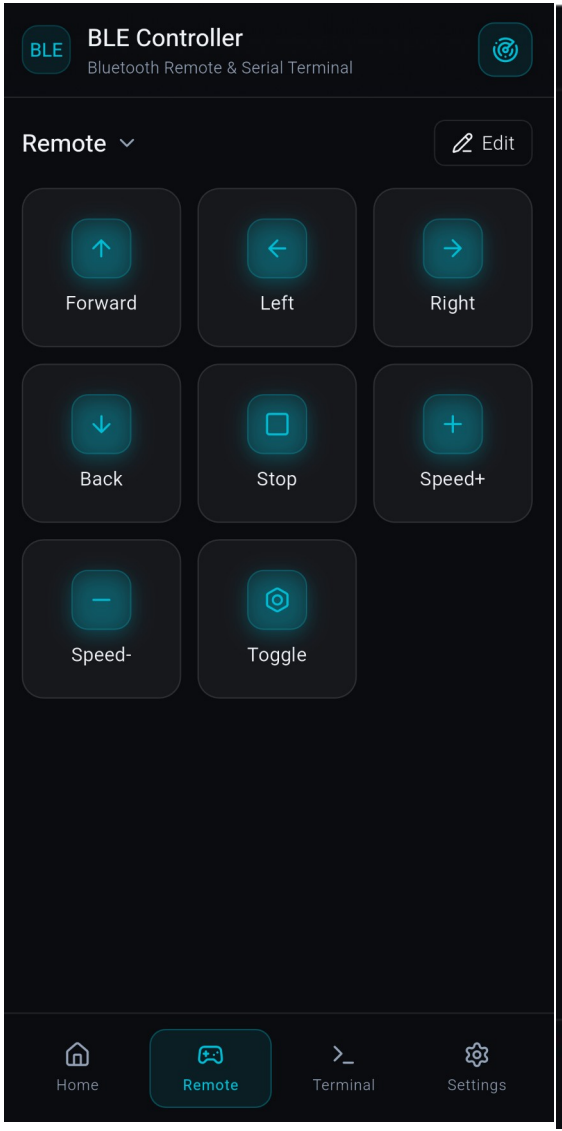
BLE Controller - Arduino ESP32 by CircuitMagic Inc.

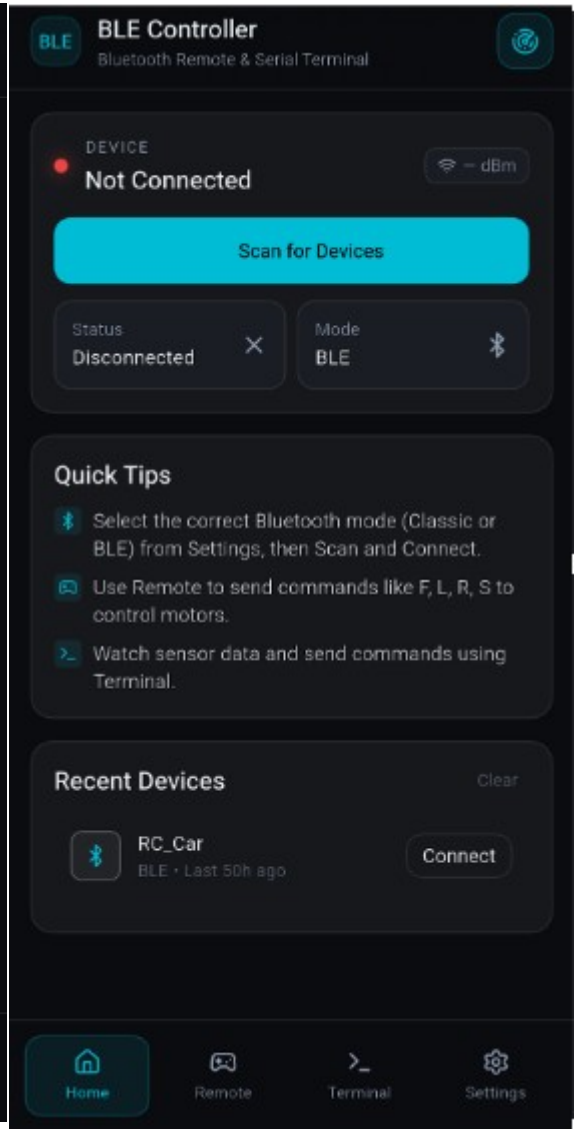
Functionality

Allows the user to create custom buttons and map them to BLE characteristics.

Command Mapping

- **'F' / 'B'**: Forward and Backward.
- **'L' / 'R'**: Left and Right turns.
- **'+' / '-'**: Speed adjustment.
- **'T'**: Toggles between driving modes.


BLE
BLE Controller
Bluetooth Remote & Serial Terminal
Remote
Edit
Forward
Left
Right
Back
Stop
Speed+
Speed-
Toggle
Home
Remote
Terminal
Settings


BLE
BLE Controller
Bluetooth Remote & Serial Terminal
DEVICE
Not Connected
– dBm
Scan for Devices
Status
Disconnected
Mode
BLE
Quick Tips
Select the correct Bluetooth mode (Classic or BLE) from Settings, then Scan and Connect.
Use Remote to send commands like F, L, R, S to control motors.
Watch sensor data and send commands using Terminal.
Recent Devices
Clear
RC_Car
BLE • Last 50h ago
Connect
Home
Remote
Terminal
Settings

Active Safety & AEB

Collision Prevention:

Uses "Active Sonar" to monitor surroundings.

Automatic Emergency Braking (AEB):

If an obstacle is detected within 15 cm, the car stops immediately and overrides user commands.

Automatic Slowdown:

Speed is automatically capped if an obstacle is detected within 40 cm.

Dual Driving Modes

One-Press Mode (Default)

A single command triggers persistent movement until a "Stop" command or new direction is received.

Hold-to-Move

- Simulates a "dead-man's switch".
- Requires a constant stream of BLE commands; if no command is received within 300ms, the car stops automatically.

Safety Sync

Motors stop immediately when switching modes via the 'T' command.

Visual Feedback

Display

Uses the built-in 12x8 LED matrix on the UNO R4.

Indications

- **Arrows**: Show direction of movement.
- **Icons**: Stop symbols and mode indicators.

Technical Implementation

Bitmap data is stored in flash memory and copied to RAM for memory-safe display.